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A spatiotemporal wind power forecasting method based on dual-view graph fusion and dual-granularity residual learning

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ABSTRACT

Accurate wind power forecasting for turbine clusters is critical for reliable power system operation. Existing spatiotemporal graph neural network (ST-GNN) methods typically rely on a single static graph representation, which limits their ability to capture diverse spatial relationships. To address these issues, we propose a Spatiotemporal Dual-View Graph Network (ST-DVGN) for wind power prediction, which models turbine interactions from two complementary views: physical spatial proximity and historical power statistical dependency. Instead of introducing complex dynamic or heterogeneous graphs, the proposed framework constructs dual-view graph representations over a fixed topology and integrates them through a unified fusion mechanism. In addition, a dual-granularity temporal module is designed to capture both local fluctuations and long-term trends in wind power time series. Extensive experiments demonstrate that ST-DVGN consistently outperforms baselines in prediction accuracy. Moreover, under cluster-level incomplete observation scenarios, the RMSE of ST-DVGN decreases by a maximum of only 6%, with the model still maintaining high prediction stability and robustness. These results indicate that dual-view spatial modeling provides an effective and robust solution for wind power forecasting in realistic operating environments.

1. Introduction

With the growing demand for wind energy, the penetration of wind power in modern power grids has increased steadily. According to the Global Wind Energy Council (GWEC), global installed wind power capacity increased by 117 GW in 2024 [Global Wind Energy Council, 2025]. However, wind power generation is inherently intermittent and volatile, as it is influenced by multiple unstable factors such as wind speed and wind direction [Holttinen et al., 2009]. Such variability imposes additional economic burdens on power dispatch and increases operational stress on power systems [Sati et al., 2025]. To mitigate the intermittency and variability of wind power, it is crucial to develop accurate and robust wind power forecasting methods. These methods can improve the utilization efficiency of wind resources and provide effective decision support for wind power dispatch, thereby facilitating large-scale grid integration.

Wind power forecasting approaches can be broadly categorized into four types: physical methods, statistical methods, traditional machine learning methods, and deep learning methods. Among physical ap-

proaches, a representative method predicts wind farm power output by using numerical weather prediction (NWP) to simulate terrain and environmental variations within the wind farm region [Feng et al., 2010]. Although physical methods comprehensively account for the physical characteristics of wind farms and are generally more reliable for long-term forecasting [Wang et al., 2023], they heavily rely on meteorological and terrain data and exhibit limited robustness to disturbances [Huang, 2019]. In addition, commonly used statistical models, such as the autoregressive integrated moving average (ARIMA) model [Liu et al., 2021], Kalman filtering [Lio et al., 2021], and Bayesian frameworks [Zou et al., 2022], can effectively model linear dependencies and stochastic fluctuations in time series. However, they struggle to capture the complex nonlinear interactions between wind power and multiple influencing factors, including meteorological conditions [Lv et al., 2024].

Compared with physical and statistical approaches, traditional machine learning methods exhibit stronger adaptability, enhanced nonlinear modeling capability, and greater capacity for handling high-dimensional features. For example, Ref. Liu et al. [2017] optimized

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the input feature selection of a support vector machine (SVM) using orthogonal experimental design, while Ref. [Lahouar and Slama \[2017\]](#) constructed a random forest model by selecting key features, such as wind speed and wind direction, based on correlation and importance measures. Although these studies achieved favorable forecasting performance, they also revealed inherent limitations of traditional machine learning methods, namely their strong dependence on manual feature engineering. Moreover, such methods struggle to automatically extract temporal dependency patterns and spatial distribution characteristics from wind power time series.

Owing to their strong automatic feature learning capability and advantages in modeling complex nonlinear patterns, deep learning methods have been widely applied in wind power forecasting research. Among temporal modeling approaches, long short-term memory networks (LSTM) [[Cali & Sharma, 2019](#)] and Transformers [[Wang et al., 2022b](#)] have been extensively used for wind power forecasting. Ref. [Ko et al. \[2021\]](#) proposed a deep residual learning approach that integrates multi-level residual networks with DenseNet, and enhances forecasting accuracy and generalization by incorporating bidirectional LSTM (Bi-LSTM). Ref. [Nascimento et al. \[2023\]](#) introduced a Transformer-based framework combined with discrete wavelet transform, which performs multi-scale decomposition on denoised multi-source data and captures temporal correlations via attention mechanisms, thereby improving both prediction accuracy and robustness. However, the aforementioned methods model wind power solely based on the historical data of individual turbines, while neglecting the interactions among turbines [[Song et al., 2023](#)]. For instance, random fluctuations in wind speed, as well as failures or aging of turbine components, can transfer spatial correlation characteristics of variables into the temporal features of wind power, thereby affecting future forecasting results [[Hou et al., 2025](#)].

In recent years, spatio-temporal analysis of wind power forecasting has increasingly become a research frontier. The combination of convolutional neural networks (CNNs) and temporal models can effectively extract spatio-temporal features [[Abedinia et al., 2024](#)]. However, CNNs rely on grid-based convolutions, which limits their ability to handle the irregular turbine layouts and dynamic spatial dependencies inherent in wind farms.

With the development of graph neural networks (GNNs), numerous studies have demonstrated their advantages in modeling irregular topologies and dynamic relationships, both in recommendation systems [[El Alaoui et al., 2021a,b, 2022, 2024a,b, El Alaoui et al., 2025, El Alaoui et al., 2025](#)] and in the wind power domain [[Huang et al., 2024, Liu et al., 2023, 2024, Xu et al., 2024, Zhang et al., 2024](#)]. For example, Ref. [Bi et al. \[2021\]](#) modeled spatial dependencies among wind farms using graph convolutional networks (GCNs) and captured temporal dependencies with gated recurrent units (GRUs), achieving higher forecasting accuracy than traditional time-series methods. Ref. [Lv et al. \[2024\]](#) proposed a spatio-temporal attention-based graph convolutional network that jointly extracts spatio-temporal features via spatial and temporal attention mechanisms, and generates predictions through a combined framework of temporal convolutional networks (TCNs) and GCNs. Although these GNN-based spatio-temporal networks can handle complex topologies and extract high-dimensional spatial features, the use of a single static graph representation still exhibits limitations in spatial feature learning capacity and robustness. On the one hand, a single graph structure has limited representational capacity and cannot fully capture multi-scale and multi-view spatial interaction patterns in wind farms. On the other hand, a strong reliance on a single topology makes such models sensitive to missing turbine-node data. When sensor failures or communication interruptions lead to partial data loss, model robustness deteriorates significantly, making it difficult to ensure stable forecasting performance.

To address the limitations of single static graph representation learning, existing studies have explored dynamic graphs, heterogeneous graphs, and multi-graph modeling. Ref. [Wang et al. \[2022a\]](#) quanti-

fied time-varying inter-node association strengths by computing dynamic spatio-temporal correlation matrices of wind speed and power, and constructed a dynamic wind farm graph representation by integrating a hierarchical directed graph structure, thereby providing more precise structural support for wind farm cluster power forecasting. Ref. [Li et al. \[2024\]](#) designed a dynamic heterogeneous graph by introducing view attributes, which integrates spatial relational connections, adaptive node embedding learning, and heterogeneity handling. The adjacency matrix is dynamically updated to construct the heterogeneous graph, thereby enhancing representational capacity in complex scenarios. Ref. [Hou et al. \[2025\]](#) constructed multiple relation graphs based on spatial distance, Pearson correlation coefficients, and fast dynamic time warping, respectively. These graphs were fused into a unified representation via an attention mechanism and combined with a multi-graph neural network-assisted spatio-temporal fusion module and a time-frequency dual-domain Transformer, leading to improved wind power forecasting performance. Although these methods alleviate the representational limitations of single static graphs to some extent, their modeling primarily focuses on dynamic relationship variations or multi-relation fusion, often relying on a single perspective or simply and prematurely aggregating multiple perspectives into a unified graph structure. Moreover, by aggregating different relationships through a unified graph structure, these approaches struggle to preserve independent information representations from distinct structural perspectives during the modeling process.

Based on the above analysis, it remains necessary to design a spatiotemporal graph learning framework that can exploit information from different structural views, without introducing complex dynamic graph updates or heterogeneous modeling assumptions. To this end, this paper proposes a Spatiotemporal Dual-View Graph Network (ST-DVGN) for wind power forecasting. The model constructs dual-view graph representations from two views: physical spatial proximity and historical power statistical correlation, and performs collaborative modeling within a unified spatio-temporal learning framework. As a result, the proposed framework improves forecasting accuracy for turbine clusters while exhibiting stable predictive performance under non-ideal observation conditions.

The main contributions of this paper are summarized as follows:

(1) From the perspective of dual-view graph learning, the wind turbine cluster power forecasting problem is revisited. Without relying on complex dynamic graph updates or heterogeneous graph modeling assumptions, the proposed approach enhances spatio-temporal dependency modeling by collaboratively exploiting two structural perspectives—physical spatial proximity and historical power statistical correlation—thereby providing a more engineering-feasible modeling paradigm for wind farm forecasting.

(2) A spatio-temporal graph learning framework with multi-level feature attention fusion is developed, which incorporates structural perspectives based on physical spatial proximity and historical power statistical correlation into a collaborative modeling process. This framework effectively improves overall forecasting accuracy and validates the effectiveness of multi-view structural modeling in complex spatio-temporal prediction tasks.

(3) A residual-enhanced dual-granularity spatio-temporal coupled forecasting framework is proposed, which enables parallel coarse- and fine-grained temporal modeling. This design effectively balances short-term fluctuation modeling and long-term trend forecasting, while alleviating the degradation problem in deep networks.

(4) The proposed method maintains stable forecasting performance under non-ideal conditions with incomplete node observations, indicating strong robustness and practical applicability.

The remainder of this paper is organized as follows. [Section 2](#) introduces the architecture of ST-DVGN. [Section 3](#) presents experimental evaluations of its performance and robustness. [Section 4](#) concludes the paper and outlines future research directions.

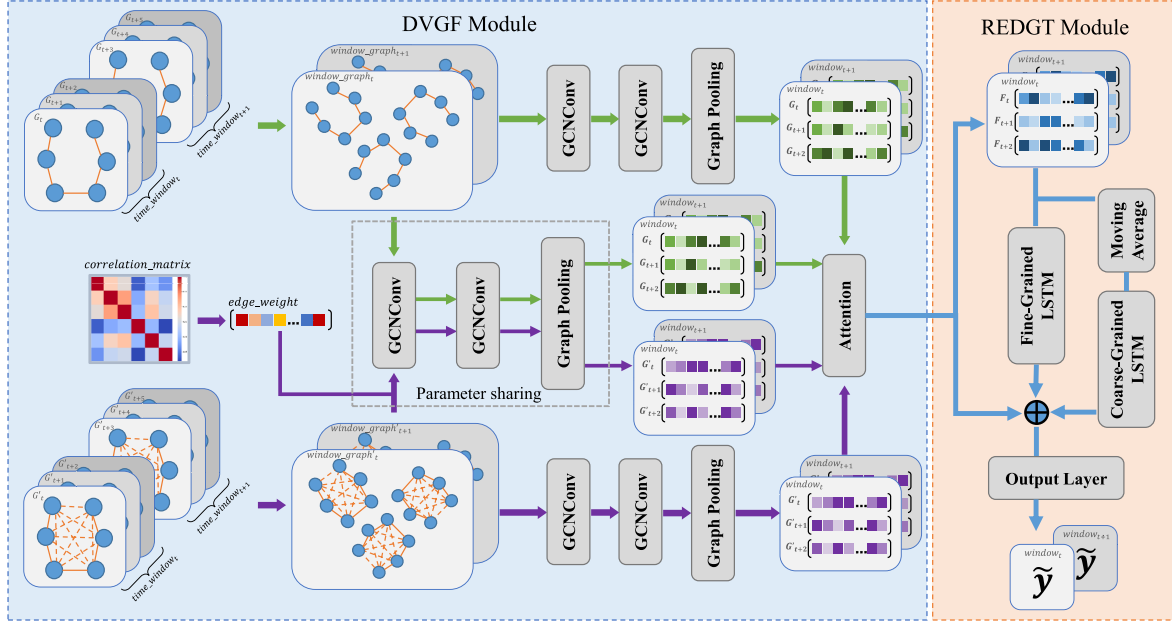


Fig. 1. Overall architecture of the ST-DVGN network model.

2. Methodology

This section introduces the overall architecture of ST-DVGN, followed by detailed descriptions of the dual-view graph fusion (DVGF) and residual-enhanced dual-granularity temporal (REDGT) modules.

2.1. Overall architecture of ST-DVGN

The overall architecture of the proposed ST-DVGN is illustrated as follows. The processing pipeline can be formally expressed as:

$$\hat{\mathbf{Y}} = F_{\text{Temporal}}(F_{\text{Spatial}}(\mathbf{X})) \quad (1)$$

Here, $\mathbf{X} \in \mathbb{R}^{N \times T \times F}$ represents the historical data of N wind turbines over T time steps with F feature dimensions. The output $\hat{\mathbf{Y}} \in \mathbb{R}^{T_{\text{future}}}$ denotes the predicted power over future time steps. The function $F_{\text{Spatial}}(\cdot)$ denotes the DVGF module, while $F_{\text{Temporal}}(\cdot)$ represents the REDGT module.

The network architecture is illustrated in Fig. 1, which consists of three core components:

(1) **Dual-View Graph Fusion Module (DVGF Module):** A physical spatial proximity graph (referred to as the **physical-view graph**) is first constructed based on the geographical locations of the wind turbines. Then, a historical power statistical correlation graph (referred to as the **statistical-view graph**) is formed by calculating pairwise correlations of historical power outputs and assigning them as edge weights in a fully connected graph. These two graphs are input into a multi-level feature extraction framework to extract spatial features.

(2) **Multi-Level Feature Attention Fusion Framework:** The dual-view graph representations are passed through their respective independent GCN layers and one shared GCN layer, resulting in four levels of spatial features. These features are then aggregated using a multi-scale attention mechanism to produce the final spatial representation.

(3) **Residual-Enhanced Dual-Granularity Temporal Module (REDGT Module):** The extracted spatial features are fed into a fine-grained LSTM to learn short- and long-term dependencies. Simultaneously, coarse-grained features are constructed via a moving average operation and processed by a separate LSTM to capture macro-level trends. Finally, the original spatial features are combined with both temporal

outputs through residual connections and passed to the output layer to generate the final prediction.

2.2. Dual-view graph fusion module

2.2.1. Construction of dual-view graph

In wind farm scenarios, interactions among turbines are primarily governed by two key factors. The first is physical spatial proximity, which reflects the interplay between geographical constraints and wake effects. The second is the statistical correlation embedded in historical power generation data, which captures operational dependencies induced by shared environmental conditions. Based on these considerations, two structured graph perspectives are constructed, namely the physical-view graph and the statistical-view graph, to represent these distinct structural views.

The physical-view graph is denoted as $\mathcal{G}_s = (\mathcal{V}, \mathcal{E}_s)$, where edges are constructed based on the Euclidean distance between turbine nodes. Specifically, a connection is established between two nodes if their distance is below a predefined threshold. The edge weights of the physical-view graph are defined as follows:

$$A_{(s,i,j)} = \begin{cases} 1, & \text{if } d_{ij} \leq \tau \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

Here, d_{ij} represents the Euclidean distance between turbine nodes i and j , and τ denotes the distance threshold.

The statistical-view graph is denoted as $\mathcal{G}_c = (\mathcal{V}, \mathcal{E}_c)$, where the edges are constructed based on the Pearson correlation coefficients calculated from the historical power output of each turbine pair. These coefficients serve as the edge weights in a fully connected graph, as defined below:

$$A_{c,i,j} = |r(\mathbf{x}_i, \mathbf{x}_j)|, \quad \forall i, j \in [1, N] \quad (3)$$

Here, $r(\cdot, \cdot)$ denotes the Pearson correlation coefficient. $\mathbf{x}_i \in \mathbb{R}^T$ represents the historical power output time series of the i th turbine node.

Fig. 2 illustrates the construction of the dual-view graph structures with turbine T0 as the central node. Fig. 2(a) shows the physical-view graph. The pink dashed circle represents the distance threshold around T0. Green solid lines indicate the connections between T0 and nearby

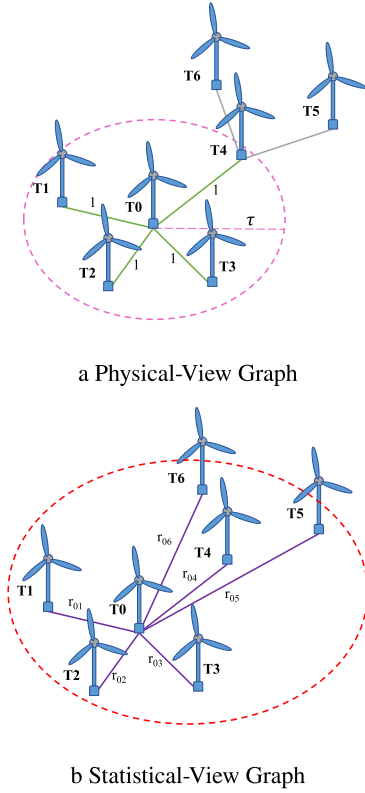


Fig. 2. Construction of the dual-view graph structures.

nodes, each assigned a weight of 1. Gray solid lines represent connections between other node pairs not involving T0, whose distances exceed the threshold. Fig. 2(b) presents the statistical-view graph. The red dashed boundary encloses all nodes, indicating that T0 is directly connected to every other node (shown by purple solid lines). Each edge weight represents the Pearson correlation coefficient between T0 and node.

2.2.2. Multi-level feature attention fusion framework

To fully exploit spatial features from the dual-view perspectives and enable collaborative representation of wind farm spatial characteristics, a multi-level feature attention fusion framework is designed to extract and integrate spatial features of wind turbines across four scales. The architecture is illustrated in Fig. 3.

We adopt GCNs to aggregate inter-turbine information through the dual-view graphs for spatial feature extraction, as GCNs provide a standard and efficient approach for modeling structured spatial correlations [Wu et al., 2021]. Graph convolutional networks can be categorized into spatial-domain and spectral-domain approaches. The GCN employed in this work is a spatial-domain convolution that serves as a simplified and approximated formulation of spectral graph convolution [Kipf & Welling, 2017]. The propagation rule is defined as follows:

$$\mathbf{H}^{(l+1)} = \sigma(\hat{\mathbf{D}}^{-1/2} \hat{\mathbf{A}} \hat{\mathbf{D}}^{-1/2} \mathbf{H}^{(l)} \mathbf{W}^{(l)}) \quad (4)$$

where $\mathbf{H}^{(l)} \in \mathbb{R}^{N \times F}$ is the node feature matrix at the l th layer, $\hat{\mathbf{A}} \in \mathbb{R}^{N \times N}$ represents the adjacency matrix with self-loops, $\hat{\mathbf{D}} \in \mathbb{R}^{N \times N}$ is the corresponding degree matrix with self-loops, $\mathbf{W}^{(l)} \in \mathbb{R}^{F \times F'}$ is the trainable weight matrix at the l th layer, and $\sigma(\cdot)$ denotes a nonlinear activation function.

To fully extract spatial information from each view and avoid information dilution and representational constraints caused by premature fusion, the dual-view graphs are processed through independent GCN

layers. Within a temporal window, at time steps $t \in [1, T]$:

$$\mathbf{H}_{s_ind}^{(t)} = \text{GCN}_{\theta_1}(\mathbf{X}, \mathbf{A}_s) \quad (5)$$

$$\mathbf{H}_{c_ind}^{(t)} = \text{GCN}_{\theta_2}(\mathbf{X}, \mathbf{A}_c) \quad (6)$$

Here, $\mathbf{H}_{s_ind}^{(t)}$ and $\mathbf{H}_{c_ind}^{(t)}$ denote the spatial features independently extracted from the physical- and statistical-view graphs, respectively. θ_1 and θ_2 are learnable parameters, while $\mathbf{A}_s$ and $\mathbf{A}_c$ represent their respective adjacency matrices.

To enable the model to explicitly learn cross-view consistent spatial representations while preserving the distinct characteristics of the physical and statistical views, it is necessary to design an explicit alignment mechanism that constrains the two views to be projected into a common representation space. Thus a shared GCN layer is introduced to jointly process both view-specific graphs. This design alleviates view-specific bias, enhances feature alignment across views, and improves model robustness, thereby promoting more stable generalization under anomalous conditions. The shared GCN layer is formulated as follows:

$$\mathbf{H}_{s_shared}^{(t)} = \text{GCN}_{\phi}(\mathbf{X}, \mathbf{A}_s) \quad (7)$$

$$\mathbf{H}_{c_shared}^{(t)} = \text{GCN}_{\phi}(\mathbf{X}, \mathbf{A}_c) \quad (8)$$

$\mathbf{H}_{s_shared}^{(t)}$ and $\mathbf{H}_{c_shared}^{(t)}$ are the spatial features derived from the shared GCN layer for physical- and statistical-view graphs, respectively. ϕ denotes the shared learnable parameters.

This results in four levels of spatial features, denoted as $\mathbf{H}_i^{(t)}$ where $i \in \{1, 2, 3, 4\}$. Each is subjected to graph average pooling to obtain compact vector representations:

$$\mathbf{h}_i^{(t)} = \text{GraphAvgPool}(\mathbf{H}_i^{(t)}), \quad i \in \{1, 2, 3, 4\} \quad (9)$$

Although the independent and shared GCNs partially overlap in terms of information content, this form of structural redundancy enables the model to retain both view-specific representations and cross-view stable representations. To prevent unconditional aggregation of redundant information that could amplify noise in the data, an attention mechanism is introduced to fuse spatial features across four scales. This mechanism allows the model to automatically perform feature selection and redundancy suppression in a data-driven manner, thereby avoiding ineffective accumulation of redundant features and further enhancing robustness under complex and anomalous scenarios. The attention mechanism takes spatial features at four scales as input, computes their respective weights through independent linear layers, and finally aggregates all spatial features via weighted summation:

$$\mathbf{z}_k^{(t)} = \mathbf{w}_k^T \mathbf{h}_k^{(t)} + b_k \quad (10)$$

$$\alpha_k^{(t)} = \frac{\exp(\mathbf{z}_k^{(t)})}{\sum_{i=1}^4 \exp(\mathbf{z}_i^{(t)})} \quad (11)$$

$$\mathbf{s}^{(t)} = \sum_{i=1}^4 \alpha_i^{(t)} \mathbf{h}_i^{(t)} \quad (12)$$

where $\mathbf{w}_k \in \mathbb{R}^d$ are trainable weights and $b_k \in \mathbb{R}$ are bias terms. The resulting vector $\mathbf{s}^{(t)}$ represents the fused spatial feature at time step t .

By concatenating the vectors $\mathbf{s}^{(t)}$ across the entire time window, the final output of the DVGF module is obtained as:

$$\mathbf{S} = [\mathbf{s}^{(1)}; \mathbf{s}^{(2)}; \dots; \mathbf{s}^{(T)}] \in \mathbb{R}^{T \times d} \quad (13)$$

2.3. Residual-enhanced dual-granularity temporal module

In multivariate wind power time series, strong non-stationarity and high noise levels are commonly observed, such as high-frequency disturbances induced by gusts. Although spatial modeling alleviates part of the raw noise to some extent, the resulting spatial feature sequences still exhibit pronounced non-stationary characteristics. To accurately capture temporal patterns in spatial feature sequences and prevent rapidly varying spatial responses from dominating backpropagation and obscuring

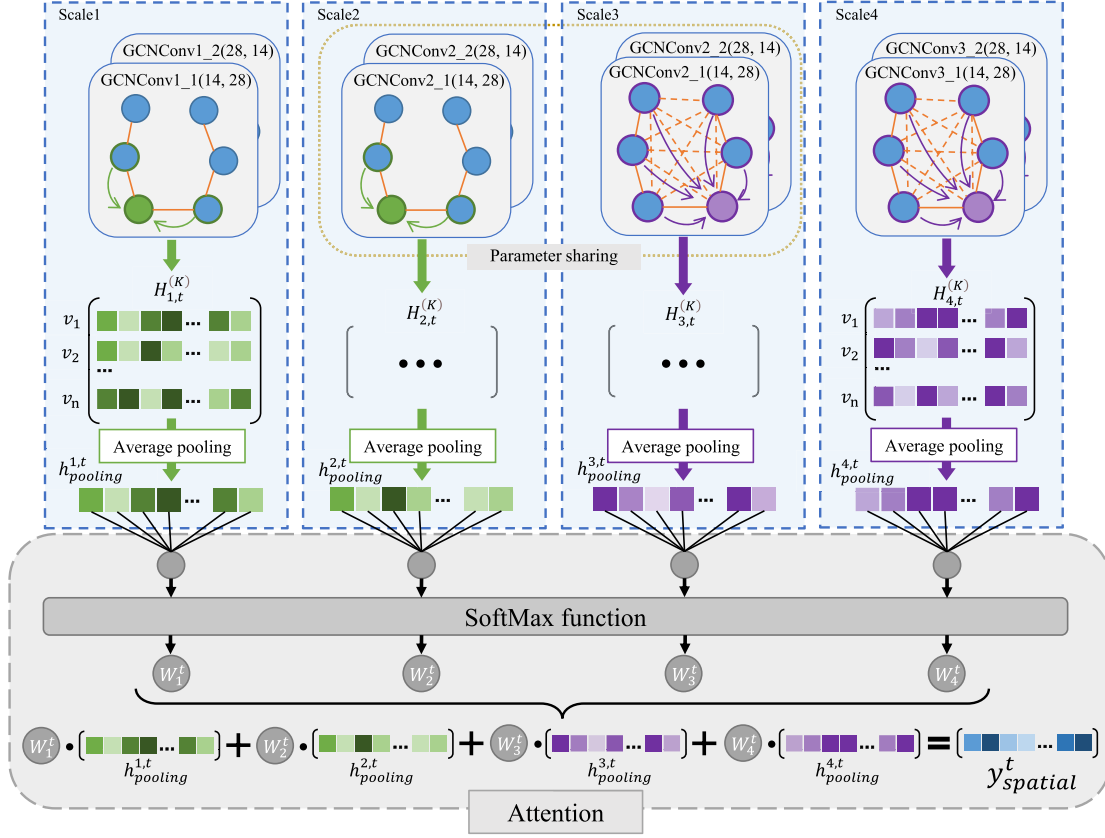


Fig. 3. Multi-level feature attention fusion framework.

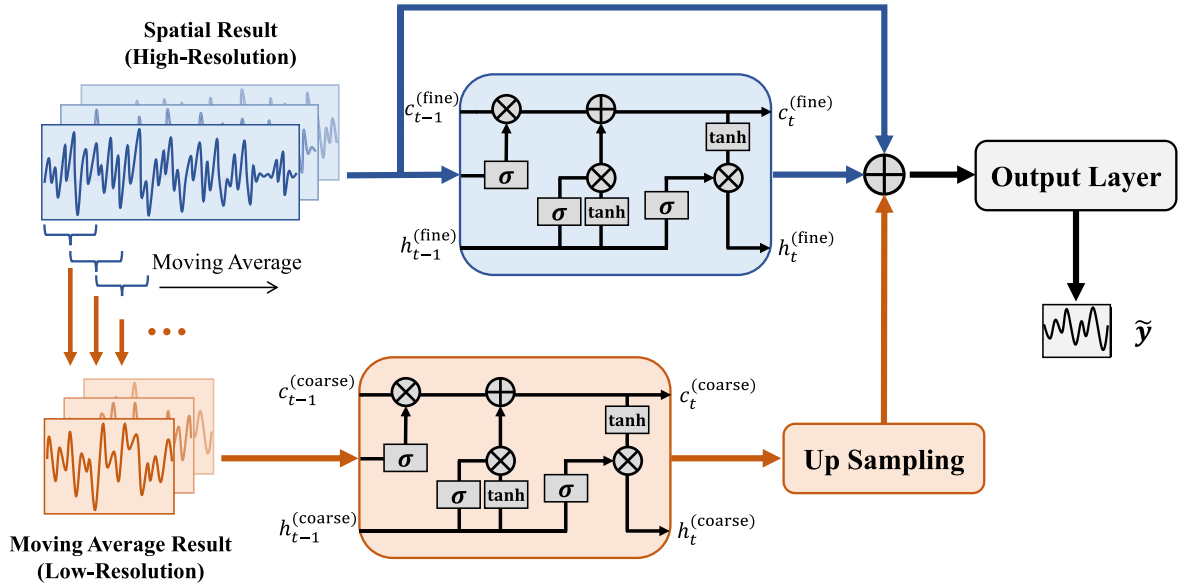


Fig. 4. Residual-enhanced dual-granularity temporal module.

long-term trends, a REDGT module is designed. The architecture is illustrated in Fig. 4.

First, a fine-grained LSTM network is employed to capture the complete temporal dynamics of the spatial feature sequence S .

$$\mathbf{H}_{\text{fine}} = \text{LSTM}_{\text{fine}}(S) \quad (14)$$

where $\mathbf{H} \in \mathbb{R}^{T \times d}$ represents high-resolution temporal features.

To prevent rapidly varying spatial features from masking slowly varying trends during backpropagation in the fine-grained LSTM, an additional coarse-grained LSTM is employed to learn trend-level temporal characteristics of the spatial features. To obtain a coarse-grained representation of the spatial features, a sliding average with window size k

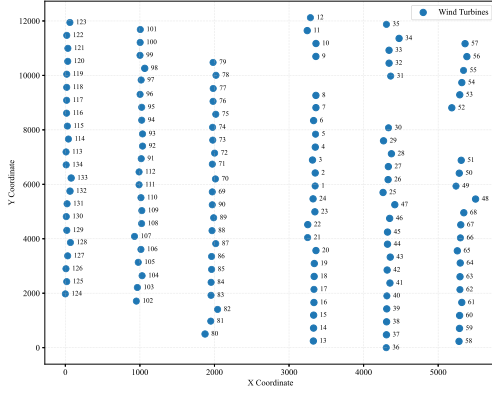


Fig. 5. Geographic location of wind turbines.

and stride s is applied to $\mathbf{S}$ to generate a coarse-grained sequence.

$$\mathbf{S}_{\text{coarse}}[t] = \frac{1}{k} \sum_{i=0}^{k-1} \mathbf{S}[t \cdot s + i], \quad t \in \{0, 1, \dots, \lfloor \frac{T-k}{s} \rfloor\} \quad (15)$$

The resulting sequence $\mathbf{S}_{\text{coarse}} \in \mathbb{R}^{(T-k+1) \times d}$ represents the trend behavior within the time window, which is then fed into a coarse-grained LSTM network to extract trend-level spatialtemporal representations:

$$\mathbf{H}_{\text{coarse}} = \text{LSTM}_{\text{coarse}}(\mathbf{S}_{\text{coarse}}) \quad (16)$$

where $\mathbf{H}_{\text{coarse}} \in \mathbb{R}^{(T-k+1) \times d}$ represents the trend features.

To align fine-grained temporal features, the coarse-grained features $\mathbf{H}_{\text{coarse}}$ are then upsampled to match the original temporal resolution T through a linear projection:

$$\tilde{\mathbf{H}}_{\text{coarse}} = \mathbf{W}_u \mathbf{H}_{\text{coarse}} + \mathbf{b}_u \quad (17)$$

where $\mathbf{W}_u \in \mathbb{R}^{T \times (T-k+1)}$ is a learnable weight matrix, and $\tilde{\mathbf{H}}_{\text{coarse}} \in \mathbb{R}^{T \times d}$ represents the upsampled trend-level features.

Finally, to mitigate potential degradation caused by network depth, the high-resolution features $\mathbf{H}_{\text{fine}}$, the trend features $\tilde{\mathbf{H}}_{\text{coarse}}$, and the original spatial features $\mathbf{S}$ are fused via element-wise addition. The fused representation is then passed through a multi-layer perceptron (MLP) to generate the final power prediction:

$$\hat{\mathbf{Y}} = \text{MLP}(\mathbf{S} + \mathbf{H}_{\text{fine}} + \tilde{\mathbf{H}}_{\text{coarse}}) \quad (18)$$

The final residual fusion mechanism further ensures that the coarse-grained pathway acts as a complementary trend correction term rather than a redundant representation of temporal features.

3. Experiments

This section evaluates the effectiveness and superiority of the proposed ST-DVGN model for wind power forecasting using the Spatial Dynamic Wind Power Forecasting (SDWPF) dataset.

3.1. Wind turbine data description

The wind turbine dataset used in this study is obtained from the SDWPF dataset. The dataset includes the spatial layout of wind turbines and environmental monitoring data collected through the Supervisory Control and Data Acquisition (SCADA) system. The SCADA system records dynamic background variables such as time, weather conditions, and internal turbine states. Data are sampled every 10 minutes from a total of 134 wind turbines deployed across the company's wind farms. The spatial distribution of the turbines is illustrated in Fig. 5.

Table 1

Comparison of ablation model performance.

Model	MAE	RMSE	R^2
Model 1	0.091	0.120	0.982
Model 2	0.090	0.126	0.980
Model 3	0.082	0.129	0.979
Model 4	0.105	0.150	0.971
Model 5	0.119	0.147	0.972
Model 6	0.064	0.103	0.986
ST-DVGN	0.052	0.088	0.990

3.2. Loss function and evaluation metrics

Considering that the SDWPF dataset is derived from real-world wind farm measurements, it inevitably contains outliers and noisy data. Therefore, this study employs the Huber loss function, which is defined as follows:

$$L_{\delta}(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2 & \text{for } |y - \hat{y}| \leq \delta, \\ \delta \left(|y - \hat{y}| - \frac{1}{2}\delta \right) & \text{otherwise,} \end{cases} \quad (19)$$

Here, y denotes the true value, $\hat{y}$ is the predicted value, and δ is a threshold parameter that determines the transition point from quadratic to linear loss. In this study, δ is set to 1.

To evaluate the accuracy of power prediction results, four standard performance metrics are employed: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2). These metrics are defined as follows:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (20)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (21)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (22)$$

where y_i is the true value and $\hat{y}_i$ is the predicted value.

3.3. Ablation experiments

To evaluate the effectiveness of the ST-DVGN and the contribution of each component, we conducted an ablation study based on single-step prediction. The following ablation variants were developed from the original model:

- **Model 1:** Removed the statistical-view graph branch (retaining the physical-view graph).
- **Model 2:** Removed the physical-view graph branch (retaining the statistical-view graph).
- **Model 3:** Removed the shared GCN (retaining the dual-view graph branches).
- **Model 4:** Removed the attention fusion mechanism and fuse spatial features of four scales via summation and averaging.
- **Model 5:** Removed the coarse-grained temporal path (retaining the fine-grained path and residual connections).
- **Model 6:** Removed the residual connections in the dual-granularity temporal module (retaining both temporal paths).

The performance of each ablated model on the test set is summarized in Table 1.

As shown in the table, removing any key component leads to a degradation in prediction performance, indicating that each module plays an essential role in the proposed framework.

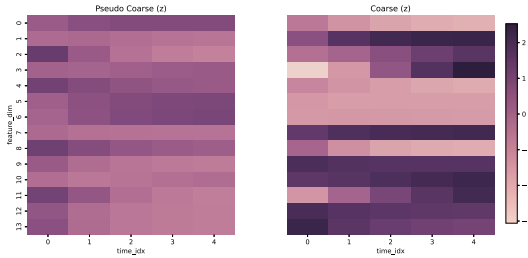


Fig. 6. Heatmaps of pseudo coarse-grained and coarse-grained temporal features.

Using a single structural view for spatial modeling (Model 1 and Model 2) results in RMSE increases of 36% and 43%, respectively, suggesting that a single-view representation is insufficient to capture the complex spatial dependencies among wind turbines. In contrast, the dual-view graph effectively integrates physical proximity and statistical correlation, yielding substantially improved performance. Moreover, removing the shared GCN layer (Model 3) causes a clear performance drop, highlighting its importance in coordinating information across views and learning shared spatial representations.

Regarding spatial feature fusion, eliminating the attention fusion mechanism in Model 4 leads to an RMSE increase of approximately 70%, indicating that naive feature aggregation is inadequate, whereas attention-based fusion is crucial for suppressing redundant information and enhancing informative features. Consequently, the attention fusion mechanism is not an instance of over-engineering but a necessary component for effective multi-view feature integration.

For temporal modeling, removing the coarse-grained LSTM in Model 5 increases RMSE by about 67%, demonstrating that fine-grained temporal modeling alone is insufficient for capturing long-term trends. The coarse-grained path provides essential macro-level temporal information. In comparison, removing the residual enhancement in Model 6 results in only a modest performance degradation of around 17%, confirming its effectiveness in alleviating degradation in deep temporal networks.

3.4. Analysis of dual-grained temporal features

The test results of the ablation model Model 5 indicate that introducing the dual-granularity LSTM contributes positively to prediction accuracy. However, performance metrics alone are insufficient to determine whether the coarse-grained temporal features effectively complement the fine-grained features in terms of temporal trend modeling. To further examine the rationality and interpretability of the proposed dual-granularity temporal modeling design, this subsection provides a qualitative analysis of the learned temporal features.

Specifically, we first extract the dual-granularity temporal features learned by the model at the epoch achieving the best validation performance. To reduce the influence of inter-sample variability, temporal features from all samples are aggregated and averaged, yielding representative fine- and coarse-grained temporal features. To enable comparison at a consistent temporal scale, the fine-grained features are processed using the same sliding-average window as the coarse-grained path, thereby constructing a pseudo coarse-grained temporal representation derived from the fine-grained features. The resulting features are visualized as heatmaps, as shown in Fig. 6.

As observed in the figure, the pseudo coarse-grained features exhibit relatively small temporal variations, reflected by smooth color transitions and the absence of extreme values in the heatmap. This suggests that temporal representations obtained by smoothing fine-grained features primarily capture local fluctuations, while exhibiting limited sensitivity to long-term global trends.

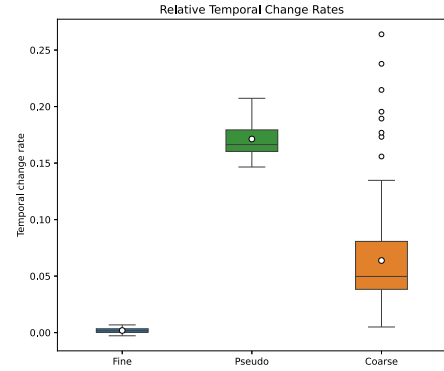


Fig. 7. Box plot of relative temporal change rate distribution.

In contrast, the coarse-grained temporal features display more pronounced dynamic variations, with clearly distinguishable extrema in the heatmap, indicating evolution over larger temporal scales. For example, the fourth dimension of the coarse-grained features shows a gradual transition from lighter to darker colors over time, revealing a clear and sustained upward trend. This observation indicates that the coarse-grained LSTM is able to capture long-term temporal dynamics that are difficult to explicitly model through the fine-grained path alone, thereby forming a complementary temporal representation.

To further examine the differences between the dual-granularity temporal features from a dynamic perspective, we analyze their relative temporal change rates and present the corresponding boxplot distributions in Fig. 7.

The fine-grained temporal features exhibit relative change rates that are highly concentrated around zero, with a narrow distribution range, suggesting that they predominantly characterize subtle variations between adjacent time steps. Although the pseudo coarse-grained features derived via sliding-average smoothing show a moderate increase in variation, their distributions remain noticeably constrained, indicating that simple temporal smoothing is insufficient to capture large-scale temporal dynamics.

By contrast, the coarse-grained temporal features present a substantially higher median and a wider distribution range of relative change rates, reflecting more significant dynamic variations over longer time horizons. These results further confirm that the coarse-grained LSTM captures long-term temporal trends through a distinct modeling mechanism, rather than acting as a redundant smoothed version of the fine-grained features. From a statistical perspective, this provides additional support for the necessity of the proposed dual-granularity temporal modeling design.

3.5. Comparative experiments

To evaluate the superiority of ST-DVGN in wind turbine power prediction, we designed a series of multi-level comparative experiments.

(1) **Baseline models:** LSTM as the temporal modeling baseline, and GCN as the spatial modeling baseline.

(2) **Advanced temporal models:**

TCN [Bai et al., 2018]: A Temporal Convolutional Network that employs dilated convolutions to capture long-range dependencies.

Informer [Zhou et al., 2023]: A Transformer-based model enhanced with a ProbSparse self-attention mechanism for efficient long-sequence modeling.

(3) **Advanced spatiotemporal models:**

CNN-LSTM [Abedinia et al., 2024]: Uses CNN to extract spatial features followed by LSTM for temporal modeling.

STGCN [Yu et al., 2018]: Models spatiotemporal dependencies using a purely convolutional graph structure.

Table 2
Model performance comparison across prediction lengths.

Prediction Length = 1			
Model	MAE	RMSE	R^2
LSTM	0.119	0.147	0.972
GCN	0.103	0.142	0.974
TCN	0.088	0.168	0.963
Informer	0.124	0.156	0.969
CNN-LSTM	0.118	0.139	0.975
STGCN	0.079	0.121	0.981
STAGCN	0.060	0.102	0.987
ENDCC-AGCN-LSTM	0.070	0.094	0.989
ST-DVGN-TCN	0.061	0.101	0.987
ST-DVGN	0.052	0.088	0.990
Prediction Length = 6			
Model	MAE	RMSE	R^2
LSTM	0.192	0.266	0.909
GCN	0.201	0.296	0.887
TCN	0.193	0.319	0.869
Informer	0.366	0.550	0.609
CNN-LSTM	0.285	0.343	0.848
STGCN	0.175	0.289	0.892
STAGCN	0.160	0.272	0.935
ENDCC-AGCN-LSTM	0.142	0.212	0.955
ST-DVGN-TCN	0.130	0.218	0.939
ST-DVGN	0.134	0.205	0.946
Prediction Length = 12			
Model	MAE	RMSE	R^2
LSTM	0.262	0.402	0.791
GCN	0.319	0.504	0.673
TCN	0.397	0.688	0.389
Informer	0.453	0.663	0.433
CNN-LSTM	0.307	0.459	0.729
STGCN	0.266	0.437	0.753
STAGCN	0.337	0.572	0.747
ENDCC-AGCN-LSTM	0.297	0.443	0.773
ST-DVGN-TCN	0.187	0.304	0.881
ST-DVGN	0.183	0.292	0.890

STAGCN [Song et al., 2023]: Employs a dual-attention mechanism to jointly learn spatial and temporal features.

ENDCC-AGCN-LSTM [Li et al., 2024]: Dynamically constructs multiple feature graph structures and applies Adaptive Graph Convolutional Networks (AGCNs) for graph switching.

(4) **Ablation-based variant**: Replacing the LSTM in ST-DVGN with TCN, denoted as **ST-DVGN-TCN**.

The experiments were designed to predict power output over three future time horizons: 10 minutes (1 time step), 1 h (6 time steps), and 2 hours (12 time steps), in order to assess the generalization performance across different forecast lengths.

As summarized in Table 2, ST-DVGN consistently achieves superior performance across both short-term and long-term forecasting horizons.

For short-term forecasting, ST-DVGN attains an RMSE of 0.088, outperforming STAGCN by 14% and the dynamic-graph-based ENDCC-AGCN-LSTM by 6%. These results indicate that jointly modeling physical spatial proximity and statistical power correlations enables more accurate characterization of local temporal fluctuations than relying on a single structural graph or predefined dynamic graph updates.

For long-term forecasting, ST-DVGN achieves an RMSE of 0.292, substantially lower than ENDCC-AGCN-LSTM, corresponding to a 34% reduction. A similar improvement is observed in MAE (38%). While ENDCC-AGCN-LSTM dynamically switches graph structures, its dependence on predefined clustered graph structures limits its flexibility in capturing evolving global dependencies. In contrast, ST-DVGN constructs the statistical-view graph directly from power correlations, allowing more adaptive modeling of long-range temporal dependencies.

Table 3
Computational complexity statistics for spatiotemporal models.

Model	Total Parameters (M)	Inference Time (ms)
STAGCN	0.062	35.968 (± 2.927)
STGCN	0.212	10.364 (± 1.651)
ST-DVGN	0.357	271.774 (± 3.010)
ST-DVGN-TCN	1.326	272.890 (± 0.697)
ENDCC-AGCN-LSTM	1.472	1883.236 (± 41.133)

This advantage is further reflected in the pronounced performance degradation of single-view graph models (e.g., STGCN and STAGCN) in long-term forecasting, whose R^2 values drop below 0.76.

To assess the temporal modeling component, ST-DVGN-TCN is compared with the original ST-DVGN in the 2-hour forecasting task. Although ST-DVGN-TCN shows a slight increase in MAE (0.187 vs. 0.183), it still significantly outperforms the standalone TCN model (MAE = 0.397). This result suggests that dual-granularity temporal modeling is critical, and that LSTM-based architectures are more effective than TCNs in capturing the non-stationary characteristics of wind power time series.

Fig. 8 presents the overall prediction curves and four representative local intervals. Across the full horizon, ST-DVGN closely tracks the ground truth, accurately capturing both peaks and troughs. In contrast, models such as Informer and CNN-LSTM exhibit noticeable deviations in multiple intervals.

The detailed regions further highlight ST-DVGN's advantages: (i) it accurately follows monotonic trends without oversmoothing, (ii) maintains stable predictions during stationary periods while suppressing noise, (iii) precisely captures peak magnitudes during rapid fluctuations, and (iv) responds promptly to abrupt changes without lag or oscillatory artifacts.

Overall, these results demonstrate that ST-DVGN achieves robust and accurate forecasting across diverse temporal regimes. Its effectiveness stems from the dual-view graph fusion mechanism and the residual-enhanced dual-granularity temporal architecture, which jointly enable comprehensive modeling of static topology and deep correlations, leading to improved accuracy, stability, and responsiveness under complex wind power conditions.

3.6. Analysis of model complexity

Based on the preceding comparative experiments, ST-DVGN has demonstrated clear advantages in prediction accuracy over existing methods. Building on these results, this subsection further evaluates ST-DVGN from the perspectives of model complexity and inference efficiency, aiming to assess its computational cost in practical deployment scenarios.

Under identical batch size settings and hardware configurations, we measure the average inference time and corresponding standard deviation over 100 test runs, and summarize the parameter counts of all models.

As shown in Table 3, ST-DVGN maintains a relatively compact parameter size of 0.357 M, which is noticeably smaller than several baseline models that incorporate deep temporal modules or complex attention mechanisms. This indicates that the proposed dual-view spatial modeling and dual-granularity temporal modeling do not incur a substantial increase in parameter complexity.

In terms of inference efficiency, ST-DVGN achieves an average inference time of approximately 272 ms. While this is higher than that of the structurally simpler ST-GCN, it remains on the same order of magnitude as ST-DVGN-TCN, which adopts a comparable temporal modeling depth, and is significantly faster than deep dynamic models such as ENDCC-AGCN-LSTM. The latter suffers from considerably higher inference time and variance due to its reliance on dynamically computing multi-dimensional correlation coefficients, constructing multiple

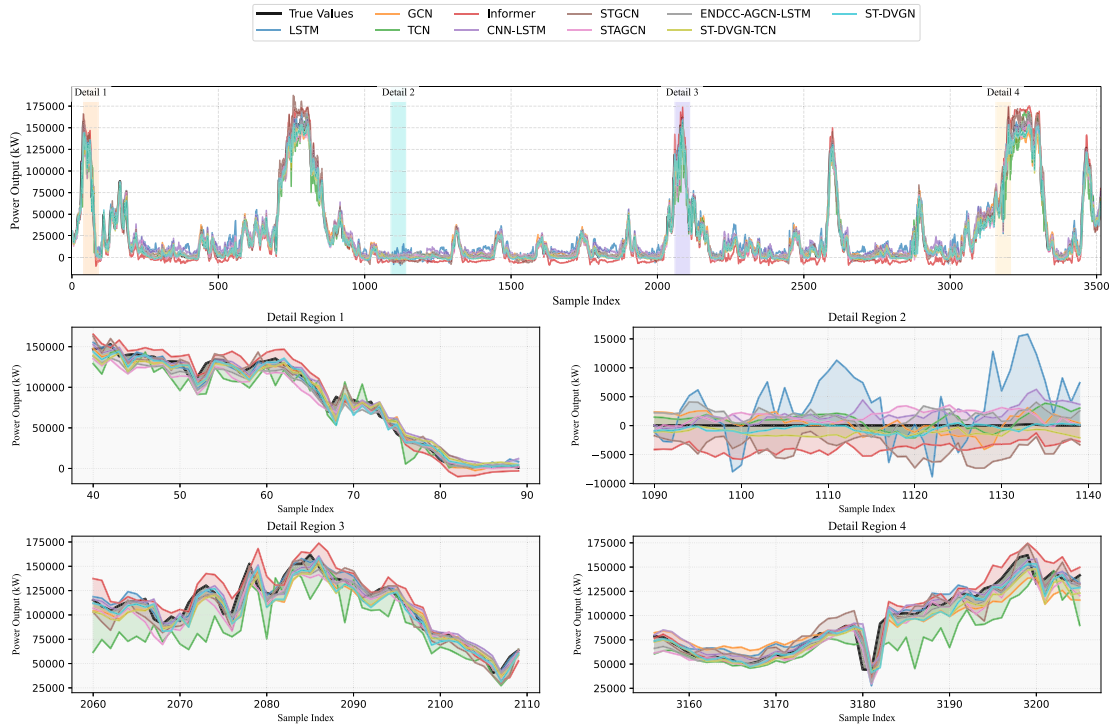


Fig. 8. Comparison of short-term forecasting models with actual values: A detailed line graph analysis across four representative intervals.

Table 4

Table of model robustness analysis results under cluster-level incomplete observation scenarios.

Models	Metric	Loss Clusters = 1		Loss Clusters = 2		Loss Clusters = 3	
		50% Loss	100% Loss	50% Loss	100% Loss	50% Loss	100% Loss
STGCN	MAE	0.080	0.081	0.081	0.082	0.083	0.086
	RMSE	0.122	0.123	0.124	0.125	0.127	0.133
	R2	0.981	0.981	0.980	0.980	0.979	0.976
STAGCN	MAE	0.061	0.061	0.063	0.064	0.065	0.069
	RMSE	0.103	0.104	0.106	0.107	0.109	0.115
	R2	0.987	0.986	0.986	0.986	0.985	0.983
ENDCC-AGCN-LSTM	MAE	0.071	0.071	0.073	0.074	0.076	0.080
	RMSE	0.095	0.096	0.099	0.100	0.103	0.110
	R2	0.988	0.988	0.987	0.987	0.986	0.984
ST-DVGN-TCN	MAE	0.063	0.064	0.065	0.066	0.067	0.070
	RMSE	0.102	0.103	0.104	0.106	0.107	0.113
	R2	0.985	0.985	0.985	0.984	0.983	0.981
ST-DVGN	MAE	0.053	0.053	0.053	0.054	0.054	0.055
	RMSE	0.090	0.090	0.091	0.092	0.093	0.095
	R2	0.988	0.988	0.988	0.988	0.987	0.986

feature-specific graph structures, and switching among multiple GCN-LSTM sub-models, combined with stacked recurrent layers and complex graph attention operations [Yang et al., 2024]. These factors collectively increase computational overhead and limit scalability in large-scale or real-time forecasting tasks.

Overall, ST-DVGN achieves a favorable balance between prediction accuracy, model complexity, and inference efficiency. The results suggest that introducing dual-view graph fusion and dual-granularity temporal modeling does not impose prohibitive computational costs, thereby supporting the practical deployability of the proposed method in real-world wind power forecasting applications.

3.7. Robustness analysis under cluster-level incomplete observation scenarios

In practical wind farm operations, data loss caused by communication failures, equipment maintenance, or shared environmental disturbances often occurs at the cluster level, resulting in incomplete observations for groups of turbines rather than isolated nodes [Pierrot & Pinson, 2025].

To realistically simulate this scenario, we design a cluster-level incomplete observation setting. Specifically, based on turbine distance information, an agglomerative hierarchical clustering algorithm is first

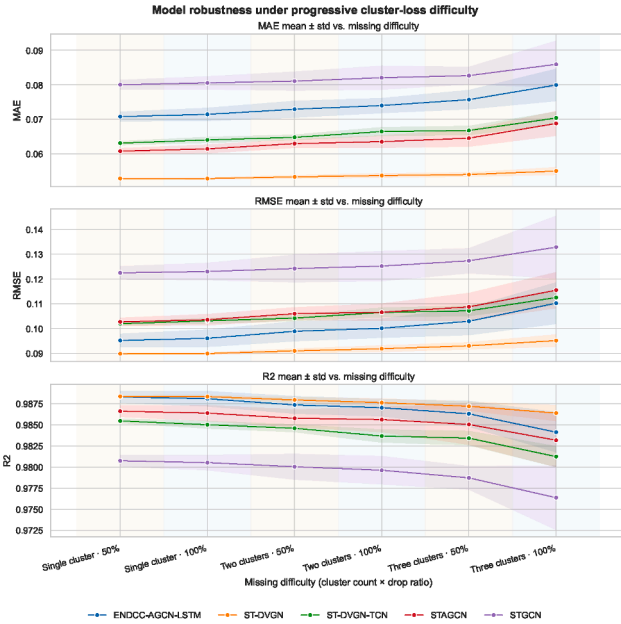


Fig. 9. Line plot of model robustness variation under cluster-level observation missing scenarios.

applied to partition the wind farm into 25 turbine clusters. Selected clusters are then subjected to two types of observation loss: partial loss (50%) and complete loss (100%) of turbine observations, while the underlying graph topology remains unchanged. In real-world systems, missing data are typically handled by null-value marking or simple interpolation strategies [Tawn et al., 2020]. Considering model compatibility across baselines, last-observation-carried-forward interpolation is adopted in this study. Finally, the severity of observation loss is progressively increased by selecting one to three clusters as missing clusters.

Under the above settings, ST-DVGN and all graph-based spatiotemporal baseline models are tasked with predicting the next-step power output. For each missing scenario, experiments are repeated 25 times, and the averaged results are reported in Table 4.

As shown in the table, ST-DVGN consistently achieves the lowest MAE and RMSE across all cluster-level incomplete observation scenarios, indicating superior robustness under non-ideal observation conditions. For instance, in the most challenging case where three clusters suffer complete data loss, ST-DVGN reduces RMSE by approximately 14% compared with the strongest baseline, ENDCC-AGCN-LSTM.

It is worth noting that ENDCC-AGCN-LSTM attains a relatively high R^2 of approximately 0.988 under the single-cluster missing scenario, approaching the performance of ST-DVGN. This can be attributed to its use of dynamic correlation coefficients (DCC) to characterize wind farm dependencies across three dimensions [Yang et al., 2024], which partially mitigates the impact of missing observations on correlation estimation—an effect conceptually similar to the role played by the dual-view graph modeling in ST-DVGN.

To further examine robustness trends, we plot the mean performance and standard deviation of all models as functions of increasing missing severity, as shown in Fig. 9.

As the number of missing clusters increases, all models exhibit performance degradation, accompanied by growing variability in MAE, RMSE, and R^2 . This indicates a progressively more challenging prediction task under severe observation loss. However, ST-DVGN shows a markedly more gradual performance decline compared with other methods, demonstrating stronger resilience to increasing levels of incomplete observation.

Moreover, ST-DVGN consistently maintains a lower standard deviation across different missing configurations, suggesting more stable pre-

dictive behavior under diverse incomplete observation patterns. These results collectively confirm that ST-DVGN offers enhanced robustness and reliability in cluster-level incomplete observation scenarios, supporting its suitability for real-world wind power forecasting applications.

4. Conclusions and future work

This paper proposed ST-DVGN, a spatiotemporal dual-view graph network for wind power prediction in turbine clusters. By modeling spatial relationships from physical proximity (physical-view) and statistical dependency perspectives (statistical-view), the proposed framework effectively captures complementary structural information without relying on complex dynamic or heterogeneous graph constructions. The dual-granularity temporal modeling further enhances the ability to learn both short-term fluctuations and long-term trends.

Experimental results demonstrate that ST-DVGN achieves superior prediction accuracy compared with state-of-the-art methods. More importantly, the proposed model exhibits strong robustness and stability under cluster-level incomplete observation scenarios, which are common in real-world wind farm operations. Computational analysis shows that these performance gains are achieved with a moderate model size and acceptable inference cost.

Based on the findings of this work, several promising directions can be naturally derived. First, the proposed ST-DVGN demonstrates stable performance under cluster-level incomplete observations while relying on fixed graph structures. This suggests that the learned dual-view spatial representations capture transferable structural patterns, motivating future exploration of transfer learning across different wind farms or operating conditions, where labeled data are limited or observation quality varies.

Second, our results indicate that effective spatial modeling can be achieved without introducing dynamically evolving or densely connected graphs. Building upon this observation, graph sparsification emerges as a meaningful extension to further reduce computational cost while preserving the most informative spatial connections, especially for large-scale wind farm deployments.

These directions directly stem from the observed robustness and efficiency of the proposed framework and aim to further enhance its scalability and applicability in real-world scenarios.

CRediT authorship contribution statement

Zhiyong Fan: Writing – review & editing, Supervision, Resources, Project administration, Methodology, Formal analysis, Conceptualization; **Zhengdong Jiang:** Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Min Xia:** Writing – review & editing, Methodology, Conceptualization; **Shuai Zhang:** Writing – review & editing, Visualization, Investigation; **Ying Yan:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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